The Relay Assignment

Measured prototype / author-review draft. Source, raw CSV, grades, event traces and verbatim prompt snapshot accompany this report. Provisional evidence grades were reviewed by the assistant, not independently hand-verified by the author.

1. What I built and why

A local Streamlit/FastAPI assistant prepares a query, retrieves/reranks paper passages, generates once with Groq GPT-OSS 120B, then releases a cited answer or refuses. Recursive chunks retain exact PDF pages and offsets. The shaping decision is to buffer the candidate for citation checks: users see progress early, but answer content arrives near completion. Citation syntax cannot prove entailment. No sign-in or corrective generation loop.

2. The real service and my stand-in

Table 1 separates published limits from observed headers; local budget refusals are not provider HTTP 429s. Groq documentation: console.groq.com/docs/rate-limits (checked November 13). The stand-in replays paired first-content/tail delays from the original tiny generic probe and adds explicit queues, faults and quotas. Its load multiplier and failure probability are assumptions; it cannot reproduce shared tenancy, real model quality or hidden provider scheduling. Larger paper-sized observations are exported separately, not pooled into the bootstrap calibration.

Table 1: limit	Value	How found	Confidence
Requests/minute	30	Official free-tier documentation	medium
Tokens/minute	8000	Documentation + response header	high
Requests/day	1000	Documentation + response header	high
Tokens/day	200000	Official free-tier documentation	medium
Requests in flight	not measured	Local cap is not provider capacity	low
Limit response	not measured	Local refusals are not upstream 429	low
Recovery time	not measured	No upstream limit/recovery observation	low

3. Results

Tables 2a/2b include every offered arrival, including failures. Percentiles use linear interpolation; small-sample p99 is descriptive, not a service guarantee. First-token latency is the runner observing committed validated answer content, not provider TTFT or browser receipt. Succeeded means completed execution, including refusals. Duration includes drain; offered/min is observed inter-arrival rate. Empty usage remains unknown. Synthetic runs use fixed retrieval evidence and therefore isolate gateway/provider capacity, not real retrieval throughput.

Table 2a scenario	Source	N / TTFT n	Duration s	Offered/min	First p50/95/99 ms	Total p50/95/99 ms
steady	real	45/13	1322.3	2.0	1918.3 / 2448.1 / 2714.7	1215.8 / 2227.7 / 2665.7
steady	stand-in	8/4	33.8	40.0	19773.2 / 23892.8 / 23981.5	11852.6 / 23749.3 / 23956.4
burst	stand-in	16/3	18.4	6251.0	17627.5 / 18310.0 / 18370.6	9335.0 / 17821.9 / 18276.0
sustained_overload	stand-in	24/2	17.0	581.3	16514.6 / 16825.8 / 16853.5	4496.4 / 14982.7 / 16706.3
per_minute_limit_halved	stand-in	12/3	17.7	296.5	16912.3 / 17290.7 / 17324.3	8162.6 / 17105.3 / 17290.3
daily_budget_exhausted	stand-in	8/0	1.8	498.8	not measured	814.4 / 1075.0 / 1120.1

Table 2b scenario	OK %	Limit %	Timeout %	Retries/req	Peak queue	Tokens/answer	USD/1000
steady (real)	77.8	22.2	0.0	0.0	not measured	3199.1	0.6
steady (stand-in)	50.0	0.0	50.0	0.0	not measured	177.0	not measured
burst (stand-in)	18.8	18.8	62.5	0.0	not measured	177.0	not measured
sustained_overload (stand-in)	8.3	50.0	41.7	0.0	not measured	177.0	not measured
per_minute_limit_halved (stand-in)	25.0	0.0	75.0	0.0	not measured	177.0	not measured
daily_budget_exhausted (stand-in)	0.0	100.0	0.0	0.0	not measured	not measured	not measured

4. Where the time goes

The slowest answered real request is decomposed below using its recorded same-process durations. Working includes retrieval and model/network time; the residual includes validation, storage and observer overhead. Almost all time elapsed before validated content. Provider queueing versus network delay cannot be separated. Request ID and raw events: evidence/TRACE.md.

Trace: retrieval	Gateway wait	Model/network	Retry	Residual	Verified first	Total ms
979.0	0.1	1308.9	0.0	498.6	2781.3	2786.6

5. When a limit is hit

The gateway reserves a conservative UTF-8-byte input bound plus completion budget before dispatch. It bounds concurrency, queued work and wait time, and persists upstream cooldowns. Near quota, optional refinement is skipped under load; no uncited weaker answer or synthetic fallback is substituted. Daily exhaustion produces a capacity outcome. Actual provider reset/recovery remains unmeasured. Peak queue is not derivable from the required CSV, so the table reports not measured; sampled queue proxies are in manifests.

Retrieval: quality versus cost

All settings hold the page-preserving recursive index at 200 embedding tokens with 30 overlap. R1 uses BM25; R2/R3 use dense + BM25 fusion and a cross-encoder. Table 3 compares different fetched-piece counts, not different chunk sizes. Keep R2 provisionally: the larger R3 context encountered local token-budget refusals. This paced shared-budget experiment is operational evidence, not an isolated statistical quality ranking. Capacity failures remain ungraded; valid citations can still support a wrong claim.

Table 3 setting	N	Correct cited %	Wrong %	Correct refusal %	Tokens/answer	First p50/95 ms	Total p50/95 ms
R1: 200 tokens, 4, no	15	13.3	6.7	33.3	4643.3	1129.5 / 1142.8	936.2 / 1276.4
R2: 200 tokens, 5, yes	15	33.3	13.3	33.3	2500.6	1918.3 / 2168.1	1762.0 / 2316.9
R3: 200 tokens, 6, yes	15	13.3	6.7	33.3	3384.7	2013.4 / 2704.5	1028.7 / 2379.7

How this library will mislead a naive retriever

Term collision: “Does dynamic chunking improve RAG retrieval?” H-Net learns sequence boundaries, not document retrieval chunks. **Split evidence:** “Why sinusoidal position encodings?” formula and extrapolation rationale span separate chunks. **Incomparable tasks:** “Is Moshi faster than the Transformer?” speech dialogue latency and machine-translation throughput are not a common benchmark.

6. Retries

Retry only transient pre-content transport/429/5xx failures, within the wait/deadline budget and a maximum of three attempts; release the concurrency slot during backoff. Never retry a partial candidate as a second answer. The controlled synthetic comparison below used identical initial-failure injection; retries improved completion but worsened tail time. An earlier random-fault comparison produced no retries and is retained locally rather than presented as causal evidence.

Policy	N	Total p50/95/99 ms	OK %	Retries/request
one_attempt	16	1797.5 / 4912.5 / 5201.0	62.5	0.0
three_attempts	16	4083.9 / 7347.8 / 8452.4	81.2	0.4

7. What the person sees

At daily exhaustion the UI shows a capacity failure, retains the request in shared local history, and does not invent an answer or reset time. A new attempt is explicit. Cancel stops local work; it cannot guarantee that already dispatched provider work is unbilled. History pagination, source excerpts and run downloads remain usable without sign-in.

8. What I'm unsure about, what I left out, what I learned

The least certain assumption is that the tiny bootstrap latency sample resembles paper-sized calls; if false, simulated overload timings are misleading. Public deployment and semantic chunking were deliberately omitted. The useful systems lesson is that adding concurrency does not add token capacity: a conservative admission bound may reject work even when observed token usage is lower. A citation checker also cannot detect a reversed causal direction. These are prototype findings, not a zero-hallucination guarantee.

Your feedback on Vaani

The author explicitly deferred website feedback until results, tables and experiments were complete. This section awaits the author's own view of what stood out, what confused them and one change they would make. It has not been replaced with the assistant's opinion. Website: https://www.vaani.dev/.

How I worked with the assistant

Author-review draft, based on the actual prompt record. Three prompts drove the work. First, asking for modular features and a project tracker constrained implementation to reviewable increments and prevented a design document from counting as a live integration. Second, asking to measure Groq before calibrating the stand-in made the simulator depend on observations rather than invented latency. Third, asking for structured event logging made source, timing, tokens and failures inspectable across the pipeline. The supplied prompt snapshot preserves the available user messages verbatim; it is not a cleaned-up reconstruction of the conversation.

The author overruled the proposed corrective RAG loop and removed sign-in, chose a Streamlit interface, selected Groq GPT-OSS 120B, and chose recursive splitting before semantic-chunking experiments. Those decisions are distinctly the author's. The author also explicitly caught the unwanted sign-in option in the proposed flow and asked for its removal. During verification, the assistant found that its first random retry comparison had produced no retries, so it retained that trial and ran an explicit failure-injection comparison. Another observed failure was an answer whose citation syntax passed but whose attention-flow direction was reversed; the provisional grade counts it wrong. The author must verify the grades and this reflection rather than submit an invented personal account of checks they did not perform.